

Artistverse

Project Plan

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Bachelor of Science

Software Engineering Program

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October 2024

A handwritten signature in blue ink, appearing to be 'Pattama Longani', is centered on the page.

Project Advisor

Asst. Prof. Pattama Longani, Ph.D.

Document History

Document Name	Version	History	Status	Date	Editable	Reviewer
Artistverse-ProjectPlan	Artistverse-ProjectPlan V.0.1	Add Chapter 1 • Project Overview • Purpose and scope • Develiable Add Chapter 2 Progress 1 • Use Case Diagram • Use Case Description Add Chapter3 • Project Team Structures • Monitoring and Controlling Add Chapter4 • ISO 29110 for Very Small Entity Add Chapter 5 •Review/Responsibility •Testing Add Chapter 6 •Schedule and Milestone Add Chapter 7 • Naming Convention •Change Management •Project Repository	Draft	08/05/2024	NS	PL

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Artistverse-ProjectPlan	Artistverse-ProjectPlan V.0.2	Edit Chapter 1 • Project Overview • Purpose and scope • Develiable Edit Chapter 2 • Development tools • Framework • Hardware and Material Resources Edit Chapter 3 • Project team structure Edit Chapter4 • ISO 29110 for Very Small Entity Edit Chapter 5 •Review/Responsibility •Testing Edit Chapter 6 •Schedule and Milestone Edit Chapter 7 • Naming Convention •Change Management •Project Repository	Edit	09/05/2024	NS,WS	PL
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Artistverse-ProjectPlan	Artistverse-ProjectPlan V.1.0	Edit Chapter 1 • Project Overview • Purpose and scope • Develiable Edit Chapter 2 • Development tools • Framework • Hardware and Material Resources Edit Chapter 3 • Project team structure Edit Chapter4 • ISO 29110 for Very Small Entity Edit Chapter 5 •Review/Responsibility •Testing Edit Chapter 6 •Schedule and Milestone Edit Chapter 7 • Naming Convention •Change Management •Project Repository	Edit	28/06/2024	NS,WS	PL
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Artistverse-ProjectPlan	Artistverse-ProjectPlan V.2.0	Edit Chapter 1 • Project Overview • Purpose and scope • Develiable Edit Chapter 2 • Development tools • Framework • Hardware and Material Resources Edit Chapter 3 • Project team structure Edit Chapter4 • ISO 29110 for Very Small Entity Edit Chapter 5 •Review/Responsibility •Testing Edit Chapter 6 •Schedule and Milestone Edit Chapter 7 • Naming Convention •Change Management •Project Repository	Publish	31/08/2024	NS,WS	PL
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Chapter 1 Introduction

1.1 Project Overview

Artistverse is platform that facilitates the songwriting process and provides an engaging experience for users. By offering a range of innovative features, Artistverse aims to make the songwriting experience easier, more enjoyable, and more efficient. Whether creating unique and interesting songs for personal enjoyment or professional purposes, Artistverse is poised to become an essential tool for music creation in today's digital era and beyond.

In progress 1, We will implement feature 1 (Dynamic Lyric Composition) that has the main features to provide users with personalized song lyrics tailored to their preferences. By integrating this feature, users will be able to easily generate lyrics based on various criteria such as mood, genre, language, and artist style. This will significantly reduce the time and effort required for users to create lyrics for their songs, allowing them to focus more on the creative aspect of music composition. Then we will implement feature 2 (Mood Detection Camera Integration) to provide users with a seamless and personalized music listening experience, allowing them to discover new songs that match their mood and preferences.

In progress 2, We will implement feature 3 (Chord progression) to assist users in understanding and implementing the selected chords, ensuring accuracy and aiding in practice and composition.

In progress 3, We will implement the last feature (AI Lyrics Analysis) to further streamline the songwriting process, providing users with comprehensive musical guidance and enhancing their creative experience.

1.2 Purpose and Scope

The purpose of Artistverse is to offer a comprehensive set of tools and features that make songwriting easier, more enjoyable, and more efficient. Whether users are creating unique and interesting songs for personal enjoyment or professional purposes, Artistverse aims to be an essential tool for music creation in today's digital era and beyond.

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1.3 Project Deliverables.

No	Deliverables/Release	Media	Copies	Date
1	Project Proposal	Document	1	26/04/2024
2	Progress1 • Project plan V.1.0 • Software Requirement Specification V.1.0 • Software Design Document V.1.0 • Test Plan V.1.0 • Test Record V.1.0 • Traceability Record V.1.0	Document	1	14/06/2024
3	Progress2 • Project plan V.2.0 • Software Requirement Specification V.2.0 • Software Design Document V.2.0 • Test Plan V.2.0 • Test Record V.2.0 • Traceability Record V.2.0	Document	1	30/08/2024
4	Progress3 • Project plan V.3.0 • Software Requirement Specification V.3.0 • Software Design Document V.3.0 • Test Plan V.3.0 • Test Record V.3.0 • Traceability Record V.3.0	Document	1	29/10/2024

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Project plan V.1.0

Software Requirement Specification V.1.0

Software Design Document V.1.0

Test Plan V.1.0

Test Record V.1.0

Traceability Record V.1.0

1.4 Acronyms and Definitions

Acronyms

UI	User Interface
SDD	Software Design Document
SRS	Software Requirements Specification
URS	User Requirement Specification
UTC	Unit Test Case
STC	System Test Case

Definition

Name	Definition
IEEE	Institute for Electrical and Electronics Engineers. Biggest global interest group for engineers of different branches and computer scientists. [1]
Plan	A document series of tasks requires meeting an objective, typically including the associated schedule, budget, resources, organizational description and work breakdown structure. [1]

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Feature	Feature Transformation of input parameters to output parameters based on a specified algorithm. It describes the functionality of the product in the language of the product. Used for requirements analysis, design, coding, testing or maintenance. [1]
Project Management	The application of knowledge, skills, tools, and techniques to project activities in order to meet or exceed stakeholder needs and expectations from a project. [1]
Project Plan	A formal, approved document used to guide both project execution and project control. The primary uses of the project plan are to document planning assumptions and the decision, to facilitate communication among stakeholders, and to document approved scope, cost, and schedule baseline. [1]
Risk	An uncertain event or condition that, if it occurs, has a positive or negative effect on a project's objectives. It is a function of the probability of occurrence of a given threat's occurrence. [1]
Risk Management	The systematic application of management policies, procedures and practices to the tasks of identifying, analyzing, evaluating, treating and monitoring risk. [1]
System Testing	Testing conducted on a complete and integrated system for evaluate the system's compliance with its specified requirements. [1]
Unit Test	A test of individual programs or modules in order to remove a design or programming errors. [1]
Traceability	The ability to trace the history, application or location of an item or activity, or work products or activities, by means of recorded identification. The establishment and maintenance of relationships between such items. Horizontal traceability describes the relationship between work products of the same type (e.g. Customer requirements). Vertical traceability describes the relationship between work products, which build or derived from each other (e.g., from customer

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	requirements to qualification test cases). Bidirectional traceability allows to directly following relationships in both directions. [1]
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Chapter 2 Infrastructure

2.1 Development tools

- Github(version 2.41.0)
- MySQL(version 8.3.0)
- VS Code (version 1.89.0)
- Pycharm (version 2023.3.4)

2.2 Framework

- Vue
- Flask

2.3 Hardware and Material Resources

- Acer Predator Helios 300 PH317-7338
 - RAM 32.00GB
 - Operating system: Windows 10 Home Single Language (64 Bit)
 - Intel(R) Core(TM) i7-10750H CPU @ 2.60GHz 2.59 GHz
- Macbook Air M2
 - RAM 8.00GB
 - Operating system: MacOS Sonoma Version 14.3.1
 - Apple M2 chip

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Chapter 3 Management Procedures

3.1 Project Team Structures

No	Participants	Role	Responsibility
1	Nichakamol Sanghong	• UI Designer • Programmer • System Analysis • Document Lead • Tester	Document: • Project proposal • Project Plan • Software Requirement Specification • Software Design Document • Test Plan • Test Record • Traceability Record Implementation • Front-end Test: • Unit Test
2	Watcharit Srichai	• Project Manager • System Analysis • Lead Programmer • Document Consultant • Tester	Document: • Project proposal • Project Plan • Software Requirement Specification • Software Design Document • Test Plan • Test Record • Traceability Record Implementation • Front-end • Backend Test: • System test
3	Asst.Prof Pattama Longani	• Project Advisor	Review and Approve: • Document • Change Request

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3.2 Monitoring and controlling Mechanisms

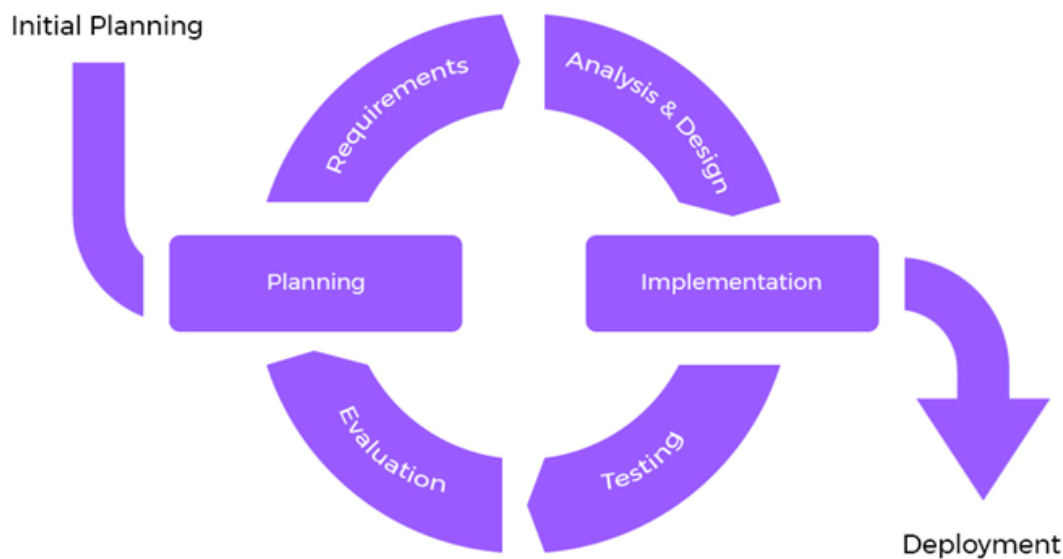


Figure 31 Iterative process

An iterative process model is a model used to develop software or software-related products. This model works in that the development process is divided into rounds, or iterations. Each round includes planning, design, development, and testing of the product. So that it can be improved and adjusted throughout development. It includes users or related parties in each round so that they can check and give suggestions for improving the product at all times. This model allows for flexibility in software development. Can be improved quickly and appropriately according to changes in needs and usage environments. [2]

3.3 Requirement Analysis

In an iterative process model, requirement analysis is an important step that helps in clearly understanding and specifying the requirements of the system or product being developed. The needs analysis team collects and analyzes data to identify user and system needs. and gain a deeper understanding of users' actual needs and usage situations. Needs analysis in an iterative process model is a process in which users and interested parties participate and provide feedback to improve product needs in every way cycle of development

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3.4 Design & Development

In the iterative process model, design and development are closely related steps. Design begins by creating a plan or model of the product according to the needs received. And development uses plans or models designed to create products. This step calls for users or interested parties to provide feedback and suggestions on product design and development.

3.5 Testing

In the iterative process model, testing is an important step used to verify the correctness and performance of the product. Each cycle of testing focuses on testing the functionality that is added to the product on a rolling basis. And this test may be a test of the entire system or only part of the system. Results from testing are used to refine and improve the product in the next round.

3.6 Implementation

In the iterative process model, implementation is the final step that involves writing code and actually deploying the designed and tested system. In this step, the developer writes code based on the design received and tests it to ensure that all functions work correctly. After completing this step, the product will be available and will be refined and improved in the next round of the iterative process model based on the feedback and needs of users or interested parties.

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Chapter 4 Quality Standard

4.1 ISO 29110 for Very Small Entity

ISO 29110 is a standard that can control the quality of the enterprise, organization, department, or project up to 25 people. The guide provides the project management process and software management process that are used to improve the quality standard of the small enterprise. The project may fulfill an external or internal contract. The internal contract between the project team and its Customer need not be explicit. [3]

4.1.1 Project Management Process

The purpose of the project management process is to establish and carry out the systematic way of the task of the software implementation project, which allows complying with the project's objectives in the expected quality, time and costs. Objectives

PM.O1. The project plan for the execution of the project is developed according to the Statement of work and validated with the customer. The tasks and resources necessary to complete the work are sized and estimated.

PM.O2. Progress of the project is monitored against the Project Plan and recorded in the Progress Status Record. Corrections to remediate problems and deviations from the plan are taken when project targets are not achieved. Appropriate treatment is taken to correct or avoid the impact of risk. Closure of the project is performed to get the customer acceptance documented in the acceptance record.

PM.O3. The Change Requests are addressed through their reception and analysis. Changes to software requirements are evaluated for cost, schedule and technical impact.

PM.O4. Review meetings with the Work Team and the Customer are held. Agreements are registered and tracked.

PM.O5. Risks are identified as they develop and during the conduct of the project.

PM.O6. A software version control strategy is developed. Items of software configuration are identified, defined and baseline. Modifications and releases of the items are controlled and made available to the customer and work team including the storage, handling and delivery of the items.

PM.O7. Software quality assurance is performed to provide assurance that work products and processes comply with the Project Plan and Requirements Specification.

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Activities:

PM.1 Project Planning Process.

PM.2 Project Plan Execution Process.

PM.3 Project Assessment and Control Process.

PM.4 Project Closure Process.

4.1.2 Software implementation process

The purpose of the software implementation process is the systematic performance of the analysis, design, construction, integration and tests activities for new or modified software products according to the specified requirements.

SI.O1. Tasks of the activities are performed through the accomplishment of the current Project Plan.

SI.O2. Software requirements are defined, analyzed for correctness and testability, approved by the customer, baseline and communicated.

SI.O3. Software architectural and detailed design is developed and baseline. It describes the software items and internal and external interfaces of them. Consistency and traceability to software requirements are established.

SI.O4. Software components defined by the design are produced. Unit test is defined and performed to verify the consistency with requirements and the design. Traceability to the requirements and design are established.

SI.O5. Software is produced performing integration of software components and verified using test cases and test procedures. Results are recorded at the test report. Defects are corrected and consistency and traceability to Software Design are established.

SI.O6. A software configuration that meets the requirements specification as agreed to with the customer, which includes user, operation and maintenance documentation is integrated, baseline and stored at the project repository. Needs for changes to the software configuration are detected and related change requests are initiated.

SI.O7. Verification and validation tasks of all required work products are performed using the defined criteria to achieve consistency among output and input

products in each activity. Defects are identified and corrected; records are stored in the verification/validation results.

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Activities:

SI.1 Software Implementation Initiation Process.

SI.2 Software Requirement Analysis Process.

SI.3 Software Architectural and Detailed Design Process.

SI.4 Software Construction Process.

SI.5 Software Integration and Tests Process.

SI.6 Software Delivery Process.

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Chapter 5 Quality Planning

5.1 Reviews / Responsibility

Stage Exit Review			
No	Stage	Review Item	Responsibility
1	Business Analysis	Project Proposal	NS, WS, PL
2	Requirement Analysis and Specification	Software Requirement Specification	NS, WS, PL
3	Project Planning	Project Plan	NS, WS, PL
4	Architecture and Detailed Design	Software Design Document	NS, WS, PL
5	Software Implementation	Source Code	NS, WS, PL
6	Unit Testing and Software Testing	Test Plan, Test Record	NS, WS, PL
7	Project Monitoring and Control	Traceability Record	NS, WS, PL

*NS=Nichakamol Sanghong

*WS=Watcharit Srichai

*PL = Asst. Prof. Pattama Longani

5.2 Testing

Stage Exit Review		
No	Test	Responsibility
1	Unit Test	NS, WS
2	System Test	NS, WS

*NS=Nichakamol Sanghong

*WS=Watcharit Srichai

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Chapter 6 Schedule and Milestones

6.1 Schedule and Milestones

Progress I: 4 May 2024 – 14 June 2024 = 42 days

Tasks	Start date	End date	Duration	Responsibility
Software requirement specification document	4 May 2024	7 May 2024	4	NS, WS
Project plan document	8 May 2024	10 May 2024	3	NS, WS
Software design document	11 May 2024	15 May 2024	5	NS, WS
Software test plan document	16 May 2024	19 May 2024	4	NS, WS
Feature #1 Dynamic Lyric Composition implement	20 May 2024	26 May 2024	7	NS, WS
Feature #1 Dynamic Lyric Composition test	27 May 2024	28 May 2024	2	NS, WS
Feature #2 Mood Detection Camera Integration implement	29 May 2024	5 June 2024	8	NS, WS
Feature #2 Mood Detection Camera Integration test	6 June 2024	7 June 2024	2	NS, WS
Prepared Progress I report and presentation	8 June 2024	14 June 2024	7	NS, WS

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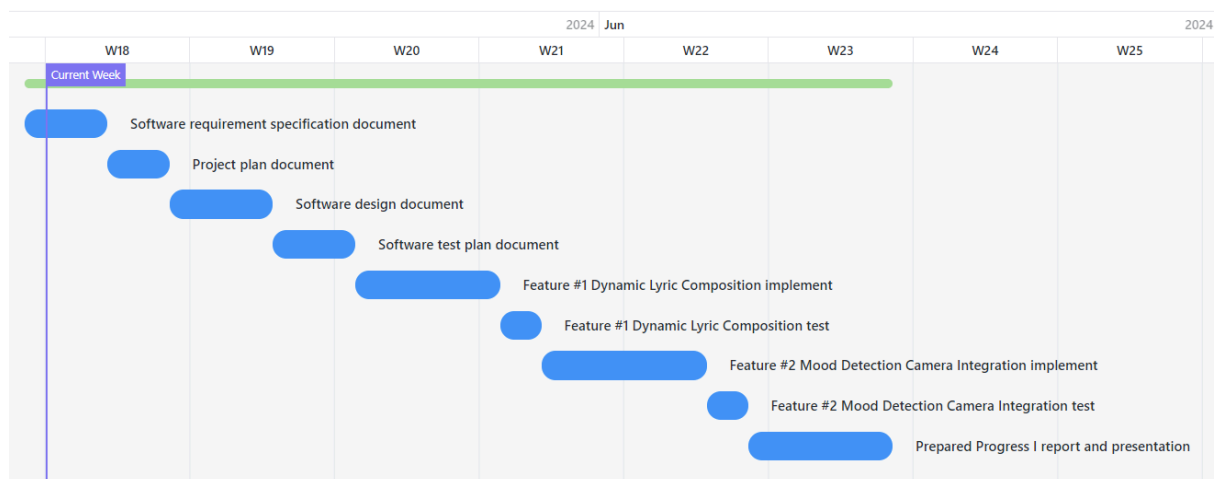


Figure 32 Progress I

Progress II: 30 June 2024 – 30 August 2024 = 62 days

Tasks	Start date	End date	Duration	Responsibility
Software requirement specification document	30 June 2024	5 July 2024	6	NS, WS
Project plan document	6 July 2024	11 July 2024	6	NS, WS
Software design document	12 July 2024	16 July 2024	5	NS, WS
Software test plan document	17 July 2024	21 July 2024	5	NS, WS
Feature #3 Chord progression implement.	22 July 2024	2 August 2024	12	NS, WS
Feature #3 Chord progression test	3 August 2024	9 August 2024	7	NS, WS

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Feature #4 AI Lyrics Analysis implement	10 July 2024	19 August 2024	10	NS, WS
Feature #4 AI Lyrics Analysis test	20 August 2024	25 August 2024	6	NS, WS
Prepared Progress II report and presentation	26 August 2024	30 August 2024	5	NS, WS

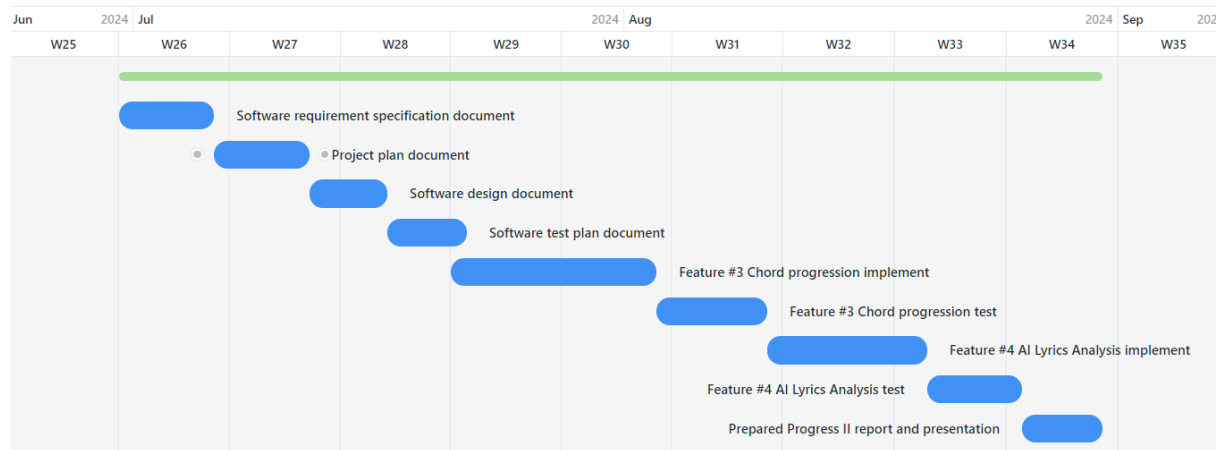


Figure 33 ProgressII

Final Progress: 25 September 2024 – 29 October 2024 = 35 days

Tasks	Start date	End date	Duration	Responsibility
Wrap up and review all of the documents	25 September 2024	2 October 2024	8	NS, WS
Prepared final report and presentation	3 October 2024	29 October 2024	27	NS, WS

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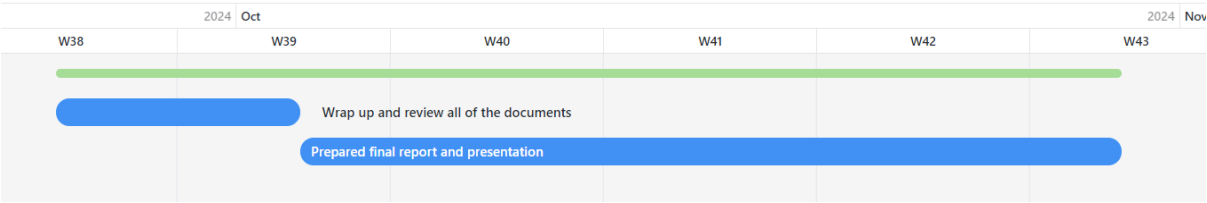


Figure 34 Final Progress

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Chapter 7 Software Configuration Management

7.1 Naming Convention

The filename format that we using for all document is Artistverse-[File Name] V.[Version].[FileType]

- File Name: The name of each process phase.
- Version: This part is the version of the file. The version number will be in this format “[Main version].[Subversion]”
- File Type - The type of the file format for example .docx or .pdf.

7.2 Change Management

Change management manages all of the changes in the project during the development process. All the change requests will be recorded into the change record document. The procedures for managing changes are:

1. Discuss with an advisor about the change.
2. Record the change information to change the document.
3. Send the change request to an advisor.
 - 3.1 Request accepted: change document and software follow the change information.
 - 3.2 Request rejected: continue and find alternative solutions.

7.3 Project Repository

- Microsoft Word: For document management
- GitHub: For source code management

Item	File Name	File Type	Baseline Version
Project Proposal	Artistverse-Proposal V.[Version]	.docx	2.0
Project Plan	Artistverse-ProjectPlan V.[Version]	.docx	1.0
Software	Artistverse-SRS V.[Version]	.docx	1.0

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Requirement Specification			
Software Design Document	Artistverse-SDD V.[Version]	.docx	1.0
Test Plan	Artistverse-TestPlan V.[Version]	.docx	1.0
Test Record	Artistverse-TestRecord V.[Version]	.docx	1.0
Traceability Record	Artistverse-TraceabilityRecord V.[Version]	.docx	1.0

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Chapter 8 Risk Management

8.1 Risk Identification and Solutions

No	Risk Statement	Solution
1	Lost the project file	Always backup the file or use cloud storage to keep the file.
2	Requirement change.	Create the change request and consult it with the advisor
3	Internet connection went down.	Try to reconnect or find another internet signal that can be used.
4	The team member is sick to the extent that he cannot work.	Reprioritize the tasks and assign them to other members.
5	Failed to meet the project schedule.	Prioritize tasks and implement the critical tasks first.
6	Other members cannot use the development tools due to some limitations.	Change the development tools

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Chapter 9 Reference

- [1]: 610.12-1990 - IEEE standard glossary of software engineering terminology (No publication date) IEEE Xplore. [Online] Available at:
<https://ieeexplore.ieee.org/document/159342?arnumber=159342> (Accessed on May 9, 2024).
- [2] RM Online Services. (No publication date). รู้จักกับ SDLC Model ที่มีการใช้งานและพัฒนาในโลกธุรกิจ. [Online]. Available at:
<https://www.rmonlineservices.com/article/13/sdlc-model-%E0%B8%84%E0%B8%B7%E0%B8%AD%E0%B8%AD%E0%B8%B0%E0%B9%84%E0%B8%A3-%E0%B8%A1%E0%B8%B5%E0%B8%82%E0%B8%B1%E0%B9%89%E0%B8%99%E0%B8%95%E0%B8%AD%E0%B8%99%E0%B8%AD%E0%B8%A2%E0%B9%88%E0%B8%B2%E0%B8%87%E0%B9%84%E0%B8%A3%E0%B8%9A%E0%B9%89%E0%B8%B2%E0%B8%87>
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